

Prey Mimic Robot

Sensors Used:

1. RGB-D/RGB Camera

Usage: For detection (person, obstacle) logic and could be used for navigation. The sensor can produce synchronized color + depth images.

Where to get: Gazebo Sim

2. Thermal Camera

Usage: it could be used to detect humans as well using heat, it might be better than using the RGB especially in forest settings.

Where to get: Gazebo Sim (They have their own thermal camera that we can just use)

3. LiDar – 2D (Fast obstacle field)

Usage: it could build the forest path quickly and could create path around bushes

Where to get: Gazebo Sim (They have their own standard LiDar sensor)

4. IMU (Inertial Measurement Unit) OPTIONAL

Usage: it could be used for more realistic movements especially when there's a sudden hits/jerks and it could be use for stabilize motion logic.

Where to get: Need to find in github, I don't think Gazebo Sim has it.

5. Contact Sensor

Usage: To simulate the hit from an airsoft bullet.

Where to get: Gazebo Sim has its own contact sim

6. Logical Audio Microphone OPTIONAL

Usage: When there's a human nearby it will detect the noise and it could move away as if the noise startled the robot.

Where to get: Gazebo Sim has the LogicalAudioSensor plugins but might need to connect to a real microphone or an audio source

Additional note

- ROS 2 bridging
- Use of actors, Gazebo has actors that act as human that could follow a scripted path
- Gazebo Fuel, use to get the assets for the forest

Risk and Constraints (Based in real life)

Sustainability: it might need to be replaced often especially since it is going to be shot a lot of time. Sustainability wise, it will be waste of materials and electronics.

Legal & Safety: In Australia, we need permission to run robots in state forest. Local regulations might be different in each area.

Cost: If we were to build it in real life it would cost more than \$100k, might not be worth it to manufacture.

Robot Shape (Simulation Constraint): Using "real" legs for the robot will be really hard, maybe we can just simulate it as a moving object but not really moving each joint of the robot's leg.

Overall in terms of constraint and risks, if we just simulate it, there would not be any problem but manufacturing it in real life might be a bit hard and might not be feasible.

ROS 2 Package

Behaviour: BehaviorTree.CPP + BehaviorTree.ROS2 (Make our robot behave like an animal with idle, alert, flee, hide)

For the gazebo sim topics: ros_gz_bridge (it is so that our simulated sensors can show up as a ROS topics)

Navigation: - Nav2 (if we want to have goals for the robot to go, if the robot wants to flee)

- Rtabmap_ros (easy RGB-D sensor to produce maps/costmaps)
- Grid_map / grid_map_ros (mutli layer 2D grid layers for elevation, roughness, traversability)

Recommended Robot

Use a wheeled base robot like Husky or Jackal.

Modification:

- Add our sensors
- Add a visual mesh (an animal costume, not a must)
- Add a dummy leg that makes it move or trot but without the physics so its just pure visual

How to make the forest

- Gazebo sim + ros_gz_bridge
- Make uneven terrain with and SDF height map (from grayscale image or DEM)
- Add trees, rocks, bushes from Gazebo Fuel (it's a drag and drop GUI or we can code it)
- Optional to make it more real: particle emitter (fog/smoke), windEffects system (wind), Actors (as mentioned before, humans).

Prey Like Behaviour Design

- Idle: head-sweep
- Alert: elevate "head" if human is in sight
- Flee: follow a path using the Nav2 then hide
- Hide: Crouch if possible for a certain amount of seconds
- Startle: if there's a sudden sound or fast object approaching.

Final Opinion

Basically it is feasible if we keep it as a ROS 2 simulation or in a controlled area and not on local forest. Might be a bit hard to implement all of the sensors but it is still do able. Most sensors are ready to use. For the robot, we can use Husky or Jackal simulators, there are many examples of it using Nav2.